Zoha Azimi

Researcher & University Assistant
Institute of Information Technology (ITEC), Universität Klagenfurt, Austria
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Google Scholar Github

Research Interests

Artificial Intelligence, LLMs, Edge AI, Video Streaming, Energy Efficiency

Education

Ph.D. in Information Technology (In Progress) Sep. 2023 – Present
Universität Klagenfurt, Austria

- Research on **AI**-assisted energy-efficient video streaming
- Supervisors: Professor Christian Timmerer, Professor Radu Prodan
- Advisor: Dr. Reza Farahani

M.Sc. in Artificial Intelligence Sep. 2021 – Jul. 2023
Alma Mater Studiorum – Università di Bologna, Italy

- Thesis: **Deep learning**-based multi-step ahead energy demand forecasting in industrial applications
- Supervisor: Dr. Michele Lombardi

M.Sc. in Electrical Engineering (Telecommunications) Sep. 2012 – Feb. 2015
Iran University of Science and Technology, Tehran, Iran

- Thesis: Nondestructive testing for defect detection of microstrip boards using near magnetic field
- Supervisor: Dr. Majid Tayarani

Experience

University Assistant Sep. 2023 – Present
Universität Klagenfurt, Austria

- **Research Focus**
  - Investigating **AI**-assisted energy-efficient video streaming
  - Exploring **LLM**s for video processing with a focus on sustainability
- **Teaching**
  - Introduction to Structured and Object-Oriented Programming: Instructed foundational programming concepts, emphasizing problem-solving and clean coding practices in Java.
  - Distributed Systems: Taught key concepts including Java RMI, socket programming, web services, distributed algorithms, peer-to-peer systems, virtualization (Docker/VMs), and orchestration (Kubernetes). This course also reflects my practical experience in designing and implementing distributed architectures in real-world scenarios.

Research and Development Specialist Feb. 2023 – Jul. 2023
Maps Group — Sharing Knowledge, Italy

- Designed and developed machine learning models for energy consumption forecasting across various industries using historical data.
- Explored and evaluated **different neural network architectures, including CNNs, LSTMs, and other deep learning models**.
- Conducted comparative analysis to identify and propose the most accurate and efficient model for each case study, based on domain-specific requirements and performance metrics.

Academic Research Projects Sep. 2021 – Jul. 2023
Alma Mater Studiorum – Università di Bologna, Italy

- **Skin Cancer Detection**: Developed a **deep learning** pipeline for skin cancer classification, including image preprocessing and model training using the ISIC dataset.
- **Breast Cancer Detection**: Applied a **Convolutional Neural Network (CNN)** to classify breast cancer subtypes from histopathological images using the BreakHis dataset.
- **Chest X-ray Image Segmentation**: Implemented a **UNet-based** model for segmenting anatomical structures in chest X-rays to assist in automated medical diagnosis.
- **Medical Abbreviation Disambiguation**: Built a **natural language processing (NLP)** pipeline leveraging BERT-based architectures to disambiguate abbreviations in clinical texts, improving understanding and accuracy in medical document analysis.

Publications

1. **Z. Azimi**, R. Farahani, R. Prodan, C. Timmerer, *ELLMPEG: An Edge-based Agentic LLM Video Processing Tool*, 17th ACM Multimedia System Conference (MMSys), 2026.
2. **Z. Azimi**, R. Farahani, V.V. Menon, C. Timmerer, *Predicting Encoding Energy from Low-Pass Anchors for Green Video Streaming*, 1st International Workshop on Intelligent and Scalable Systems Across the Computing Continuum (ScaleSys), 2025. [DOI]
3. **Z. Azimi**, R. Farahani, V.V. Menon, C. Timmerer, R. Prodan, *Towards **ML**-Driven Video Encoding Parameter Selection for Quality and Energy Optimization*, 16th International Conference on Quality of Multimedia Experience (QoMEX), 2024. [DOI]
4. **Z. Azimi**, A. Premkumar, R. Farahani, V.V. Menon, C. Timmerer, R. Prodan, *Decoding Complexity-Aware Bitrate-Ladder Estimation for Adaptive VVC Streaming*, 32nd European Signal Processing Conference (EUSIPCO), 2024 [DOI]
5. **Z. Azimi**, R. Farahani, C. Timmerer, R. Prodan, *Towards an Energy-Efficient Video Processing Tool with **LLMs***, Proceedings of the 4th Mile-High Video Conference, 2025 [DOI]
6. R. Farahani, **Z. Azimi**, C. Timmerer, R. Prodan, *Towards **AI**-assisted Sustainable Adaptive Video Streaming Systems: Tutorial and Survey*, arXiv preprint arXiv:2406.02302, 2024 [DOI]
7. **Z. Azimi**, R. Farahani, R. Prodan, C. Timmerer, *ELLMPEG: An FFmpeg-like Video Processing Tool with **Large Language Models***, Submitted to the IEEE Access journal
8. R. Farahani, **Z. Azimi**, M. Colosi, C. Timmerer, R. Prodan, *LMOrch: A QoS-Aware **LLM** Query Orchestration Service on the Edge-Cloud Continuum*, Submitted to IEEE Conference on Computer Communications (INFOCOM 2025)
9. R. Farahani, **Z. Azimi**, V.V. Menon, H. Hellwagner, R. Prodan, C. Timmerer, *DQ-Ladder: A **Deep Reinforcement Learning**-based Bitrate Ladder for Adaptive VVC Video Streaming*, Submitted to IEEE Transactions on Multimedia
10. S. Afzal, N. Mehran, **Z. Azimi**, F. Tashtarian, H. Amirpour, R. Prodan, *A Survey on Energy Consumption and Environmental Impact of Video Streaming*, arXiv preprint arXiv:2401.09854, 2024 [DOI]